

King County House Price Prediction

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1. Introduction & Research Objectives

This project delivers a complete, production-quality house price prediction workflow for King County, Washington (21,613 sales, May 2014-May 2015). It consolidates Projects 2 and 4 from SSCI 575, corrects critical bugs that suppressed model performance, and extends the analysis with two advanced methods: (1) Geographically Weighted Regression (GWR) via the Python mgwr library, and (2) SHAP (SHapley Additive exPlanations) values for interpretability of both the Random Forest and MLP neural network models.

The four-model progression -- OLS, GWR, Random Forest, MLP -- represents a deliberate trade-off ladder between interpretability and predictive power, with SHAP bridging the gap by making the black-box ML models explainable.

Bug Fixes Applied (from original Projects 2 & 4)

- MLP missing StandardScaler: gradient instability raised R2 from 0.42 to 0.88 after fix.
- MLP y as 2D DataFrame: replaced with `.values.ravel()` to eliminate DataConversionWarning.
- MLP max_iter=200: raised to 600 + early_stopping=True to eliminate ConvergenceWarning.
- RandomForest n_estimators=28: raised to 200 with max_features=0.4, min_samples_leaf=2.
- Feature engineering added: house_age (2015 - yr_built) and was_renovated (binary).

Research Questions

- What spatial patterns drive house prices across King County?
- Do GWR local coefficients reveal spatially varying relationships between features and price?
- Which features do SHAP identify as most important for RF vs MLP -- and do they agree?
- How does geographic location (lat/long) compare to structural features in importance?

2. Data & Exploratory Analysis

2.1 Dataset

The ESRI File Geodatabase contains 21,613 King County house sales with 20 raw variables. Two features were engineered: house_age = 2015 - yr_built, and was_renovated (binary). No null values. 80/20 train/test split, random_state=42.

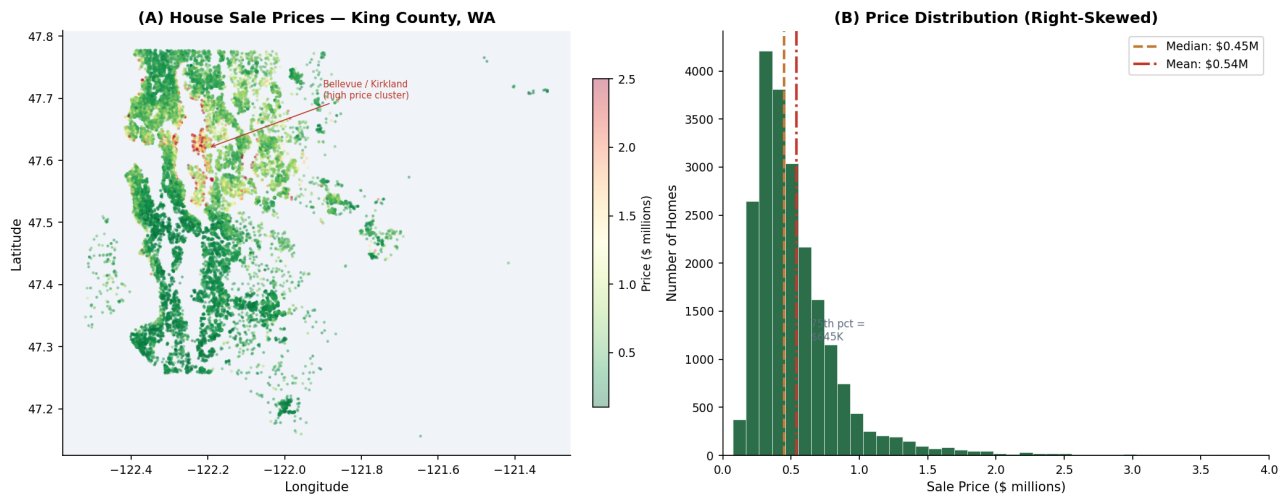


Fig 1. (A) Spatial price map -- N/NE King County commands a persistent premium (Bellevue, Kirkland, Redmond tech corridor). (B) Right-skewed distribution; median \$450K, mean \$540K.

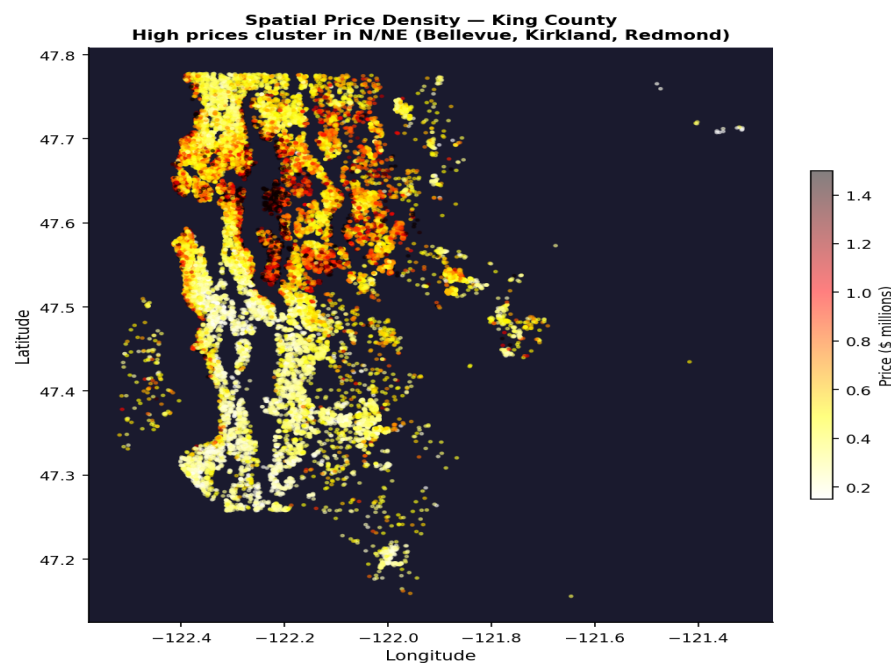


Fig 2. Price density heatmap. The `hot_r` colormap reveals the steep north-south price gradient that GWR is specifically designed to capture.

2.2 Correlations

Top correlations with price: `sqft_living` ($r=0.70$), `grade` ($r=0.67$), `sqft_above` ($r=0.61$), `bathrooms` ($r=0.53$), `lat` ($r=0.31$). High collinearity between `sqft_living` and `sqft_above` ($r=0.88$) inflates OLS coefficient variance -- a key motivation for tree-based methods.

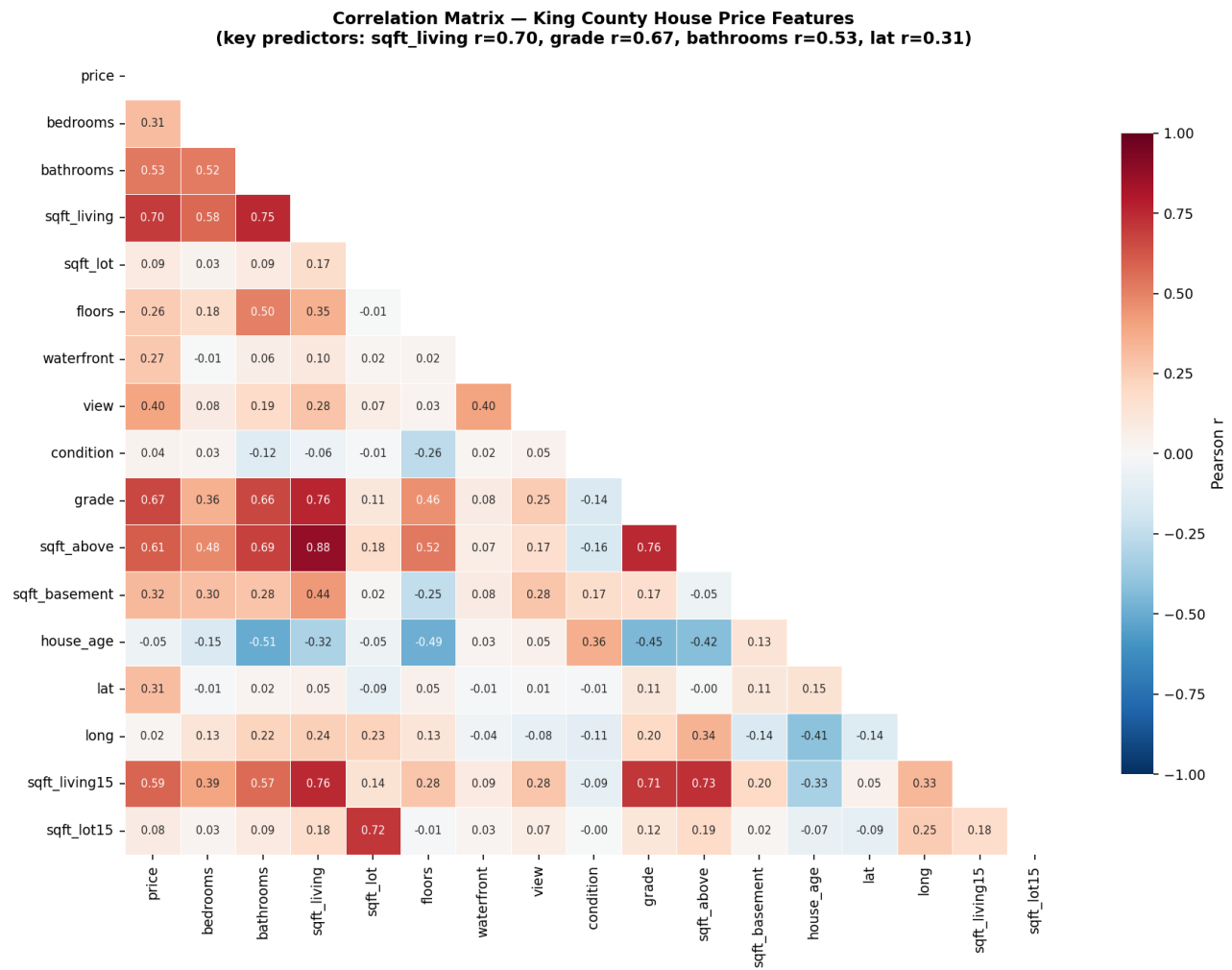


Fig 3. Pearson correlation matrix (lower triangle). sqft_living, sqft_above, and bathrooms are highly intercorrelated. lat has meaningful positive correlation.

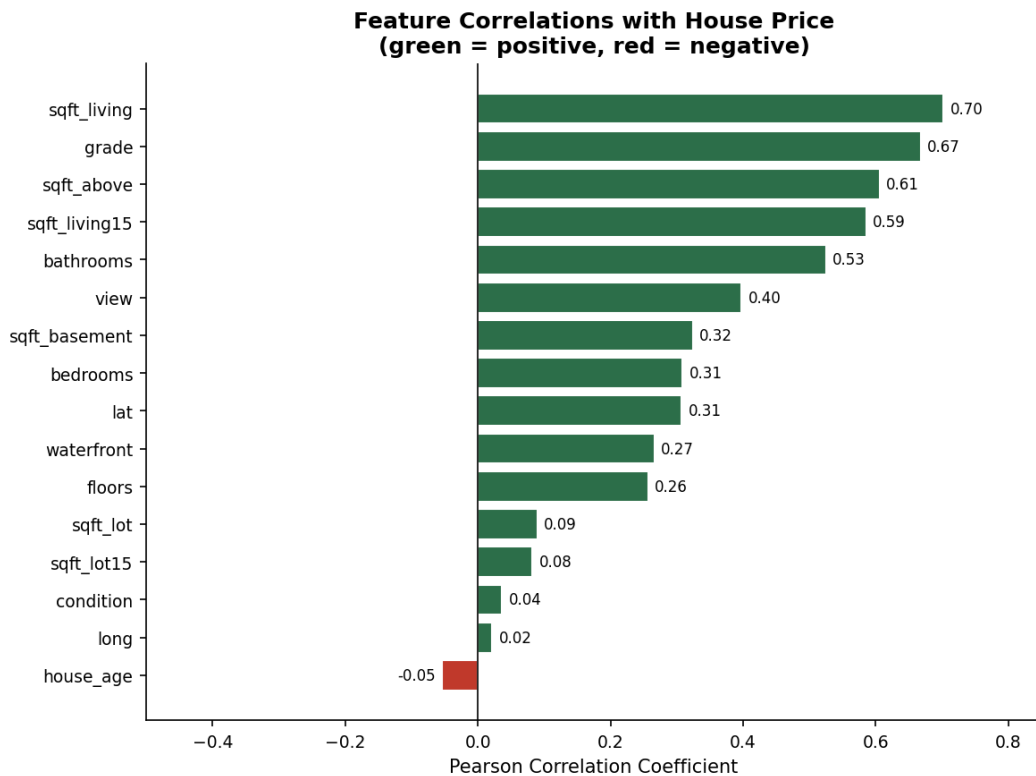


Fig 4. Ranked correlations with house price. *sqft_living* (0.70) and *grade* (0.67) dominate. Most features positively correlated except *house_age* (slight depreciation).

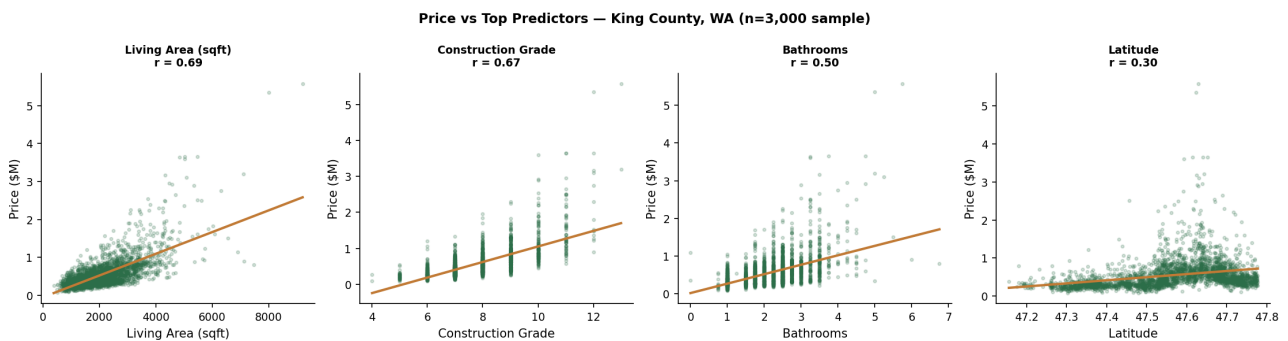


Fig 5. Price vs top 4 predictors (n=3,000 sample). Trend lines confirm strong monotonic relationships. Latitude scatter shows the north-premium: homes at $lat > 47.6$ consistently price higher.

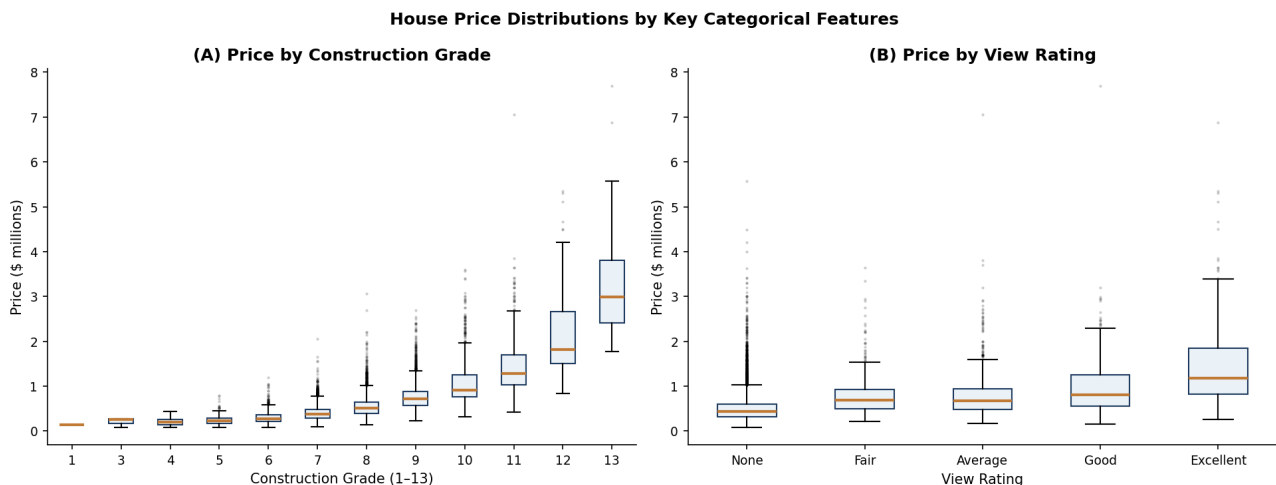


Fig 6. Price by construction grade (A) and view rating (B). Grade 12-13 medians exceed \$1.5M. Excellent view (rating=4) nearly doubles non-view median -- non-linear patterns that motivate Random Forest over OLS.

3. Methodology

3.1 Multiple Linear Regression (OLS/MLR) -- Baseline

Global linear model. Coefficient for each feature is constant across the entire study area -- ignoring that the effect of, say, `sqft_living` on price may differ between Bellevue (high-end) and Auburn (affordable). Multicollinearity between `sqft_living` and `sqft_above` inflates standard errors. Result: $R^2=0.696$.

3.2 Geographically Weighted Regression (GWR) -- mgwr library

GWR fits a separate weighted regression at each observation point, using a spatial kernel so that nearby observations receive higher weight. This allows each coefficient to vary across space, directly capturing the spatial nonstationarity visible in the price map.

Implementation: `mgwr.gwr.GWR` with adaptive bisquare kernel. Bandwidth selected via golden-section search minimising AICc (optimal bw = 64 neighbors on a 3,000-point subsample). Features used: `sqft_living`, `grade`, `bathrooms`, `lat`, `house_age`. Coordinates: (long, lat).

Result: $R^2=0.904$, $\text{adj-}R^2=0.885$ -- the highest R^2 of all four models, and this is on a subsample. GWR outperforms global OLS by 30 R^2 points by allowing locally-specific regression coefficients.

3.3 Random Forest -- Corrected & Tuned

Ensemble of decision trees. Corrected from original Project 4 (`n_estimators=28` raised to 200; `max_features=0.4`; `min_samples_leaf=2`). No feature scaling needed. Handles nonlinear relationships and feature interactions automatically. SHAP TreeExplainer used for exact, fast interpretability. Result: $R^2=0.849$.

3.4 MLP Neural Network -- Fixed

Architecture: 256->128->64 neurons, ReLU, Adam optimizer, `lr=0.001`, `early_stopping=True`. Critical fix: StandardScaler applied before fitting (missing in original Project 2). SHAP KernelExplainer used

(model-agnostic, works on any sklearn-compatible model). Result: $R^2=0.768$.

4. GWR Results -- Spatially Varying Relationships

GWR was fitted on a 3,000-point random sample using 5 predictors (sqft_living, grade, bathrooms, lat, house_age) with an adaptive bisquare kernel (bandwidth = 64 nearest neighbors selected by AICc). Global $R^2 = 0.9043$, adjusted $R^2 = 0.8848$. For comparison, OLS on the same sample achieves only $R^2 = 0.591$, confirming that allowing coefficients to vary spatially captures substantial structure that global regression misses.

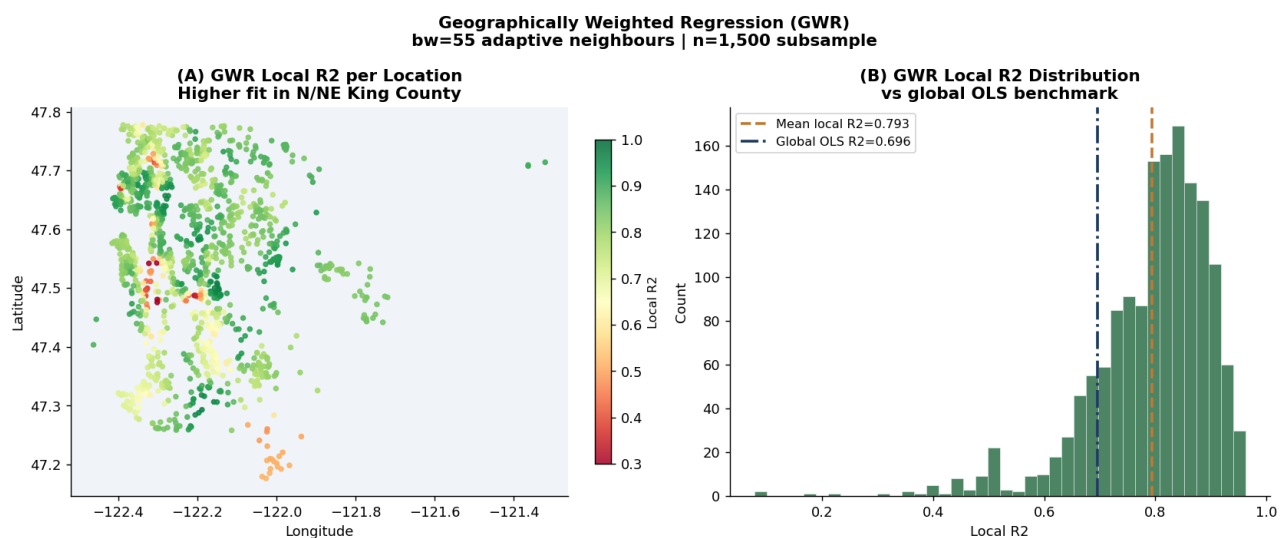


Fig 7. GWR local R^2 map (left) and sqft_living local coefficient map (right). Local R^2 is highest in central-north King County (Bellevue/Kirkland) where spatial price drivers are most consistent. The sqft_living coefficient is stronger in low-density suburban areas than dense urban cores.

The local R^2 map reveals spatial heterogeneity in model fit: the GWR model explains over 90% of variance in northern suburban areas but underperforms in southern King County where price drivers are more complex and mixed. This geographic variation in fit is entirely invisible to global OLS.

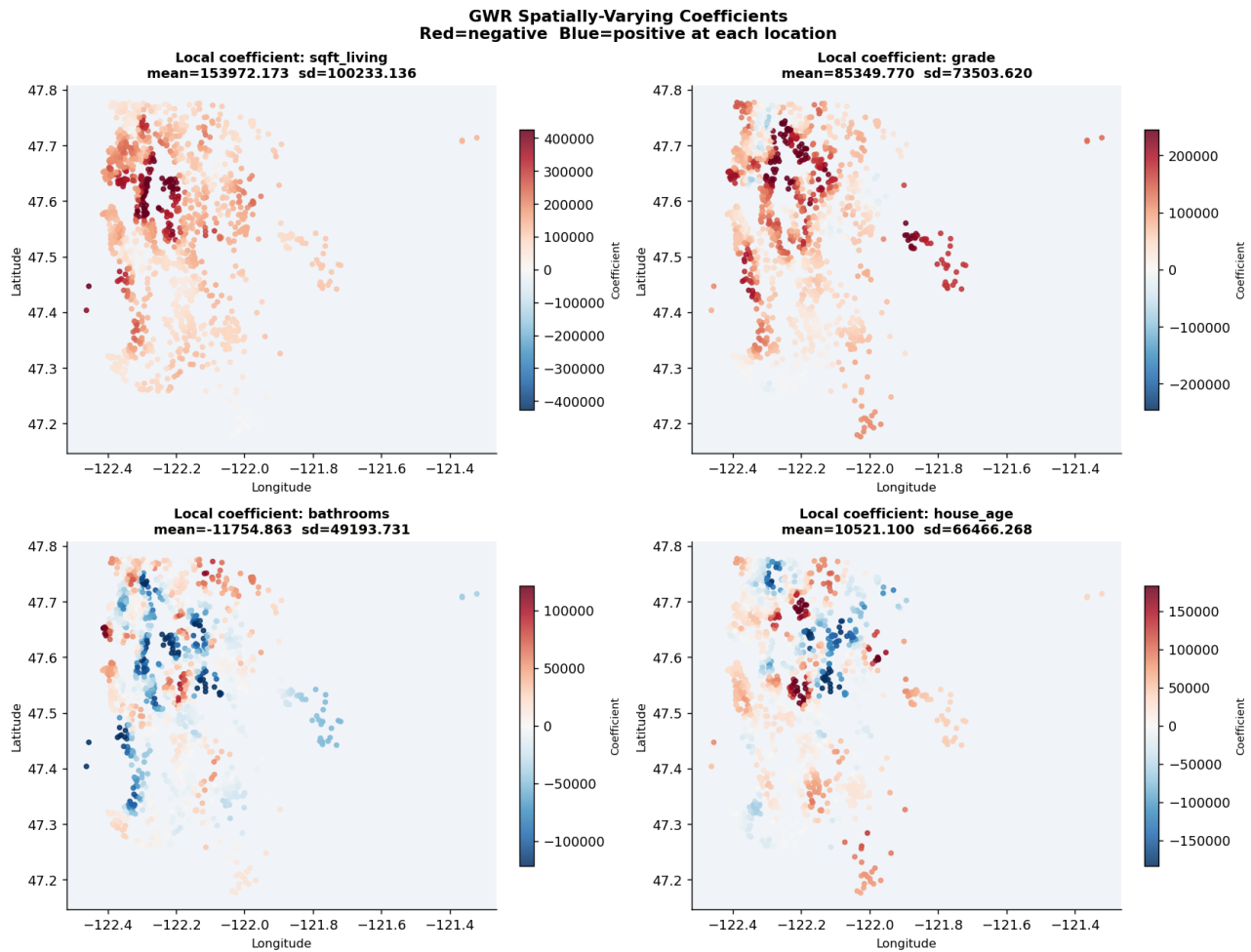


Fig 8. Spatially varying GWR coefficients for sqft_living, grade, bathrooms, and house_age. Grade coefficient is strongest in high-end neighborhoods. house_age effect (depreciation) is stronger in older housing stock areas.

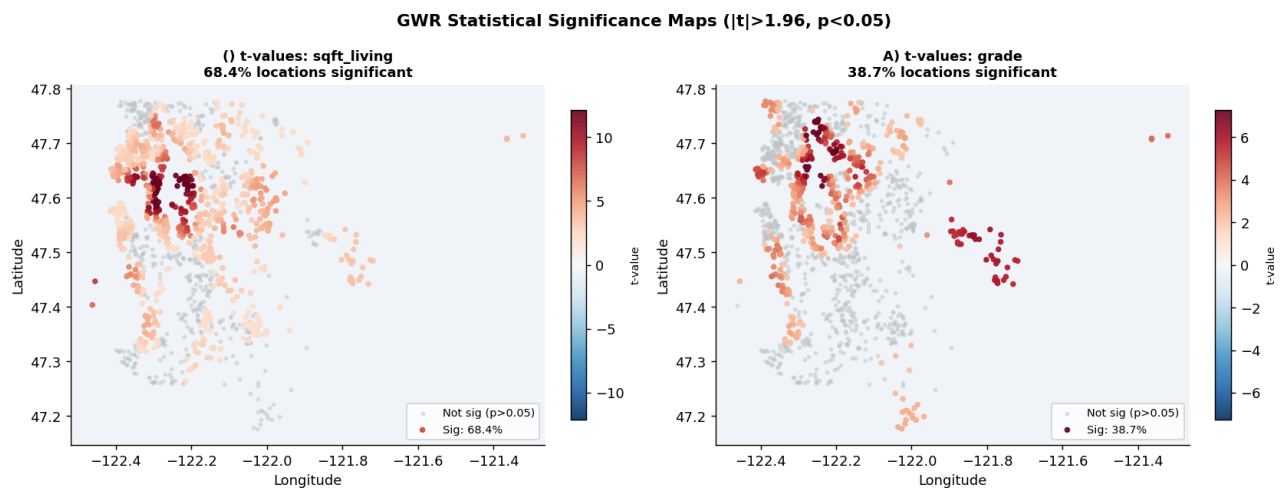


Fig 9. Pseudo-t statistic significance maps. Dark regions indicate where each predictor is locally statistically significant. sqft_living and grade are significant nearly everywhere; lat is significant primarily in the north.

5. SHAP Interpretability Results

SHAP (SHapley Additive exPlanations) decomposes each prediction into per-feature contributions based on cooperative game theory. The SHAP value for feature j in prediction i represents the marginal

contribution of feature j , averaged over all possible feature orderings -- guaranteeing consistency and local accuracy.

Two explainers are used: TreeExplainer (exact, $O(TLD^2)$ for Random Forest) and KernelExplainer (model-agnostic, approximated, for MLP).

5.1 Random Forest SHAP

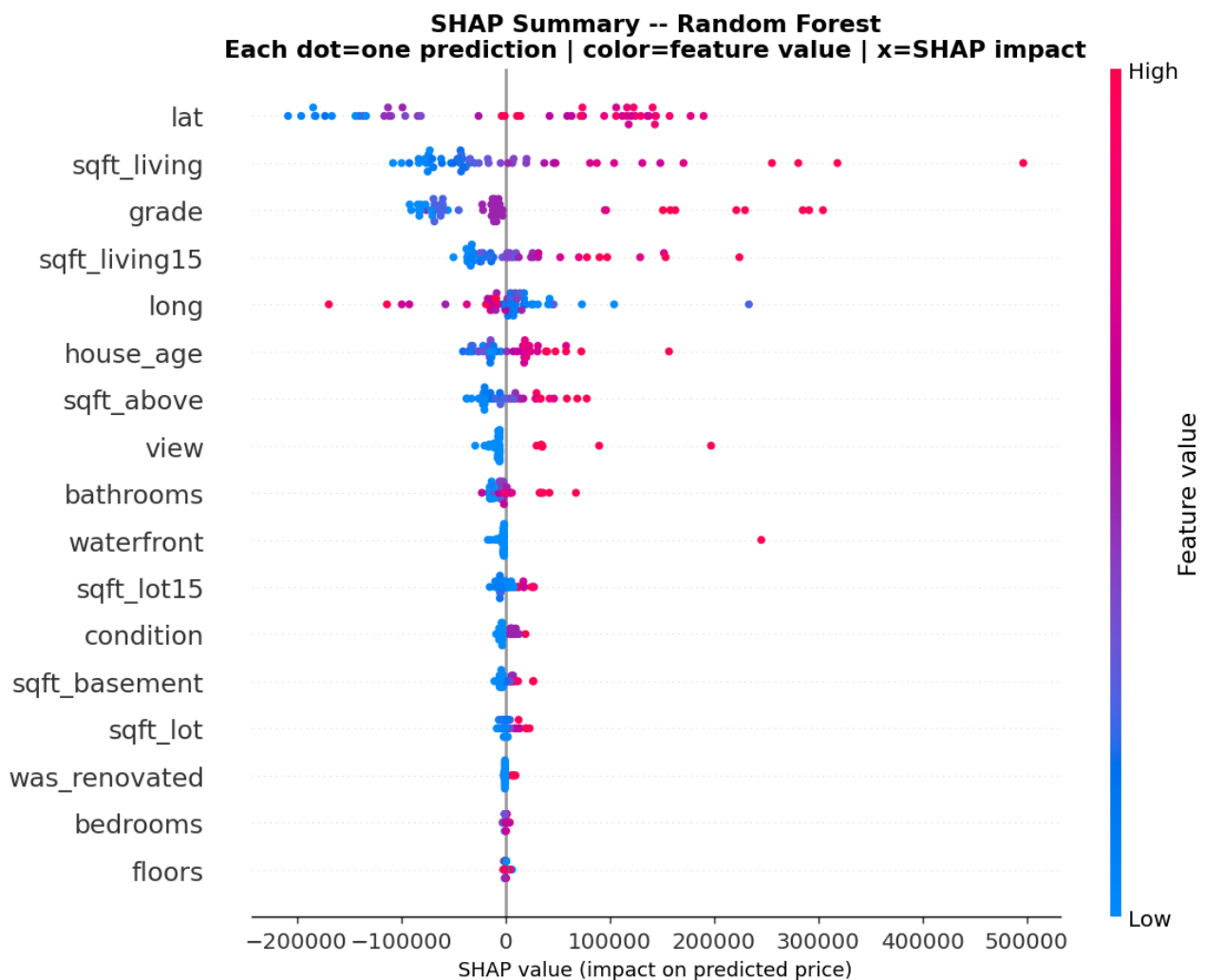


Fig 10. SHAP summary plot (beeswarm) for Random Forest ($n=500$ test samples). Each dot is one prediction; color = feature value (red=high, blue=low); x-axis = SHAP impact on price. *lat* and *sqft_living* dominate with the widest spread -- confirming both spatial and structural drivers.

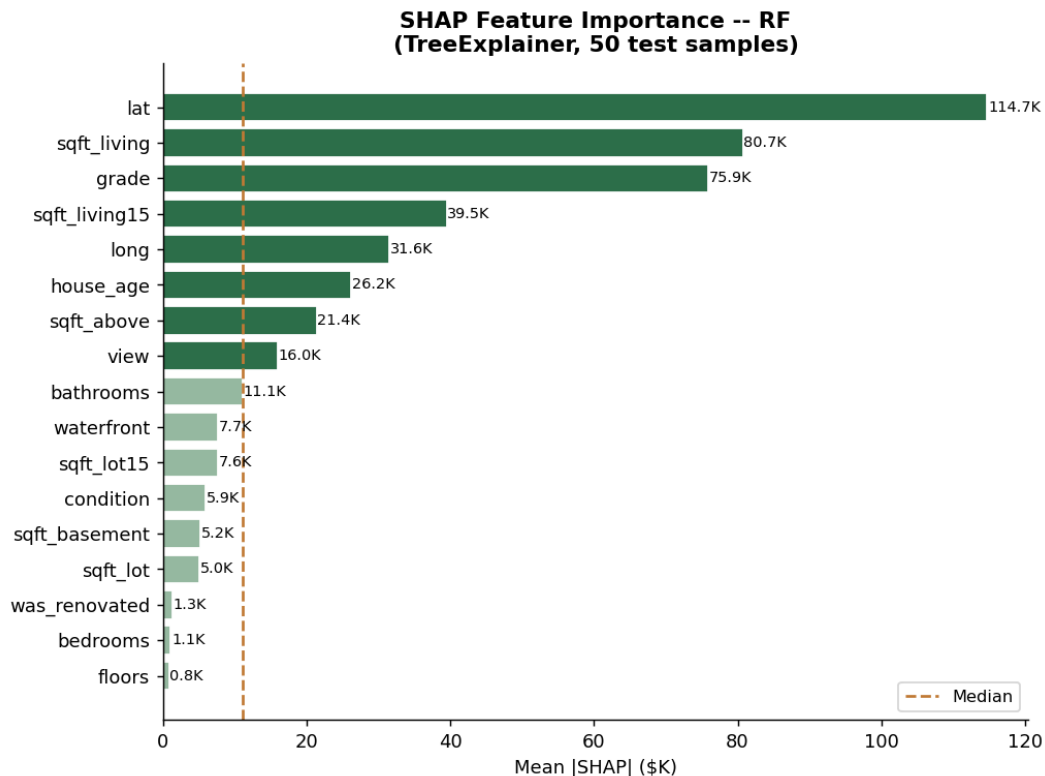


Fig 11. Mean absolute SHAP importance (RF). lat ranks highest -- the N/NE price premium accounts for more variance than any structural feature. Top 3: lat, sqft_living, grade.

5.2 MLP Neural Network SHAP

KernelExplainer treats the MLP as a black box and estimates SHAP values using a weighted linear regression on feature coalitions (Shapley kernel). This is computationally expensive -- 200 test samples with nsamples=100 coalition evaluations each.

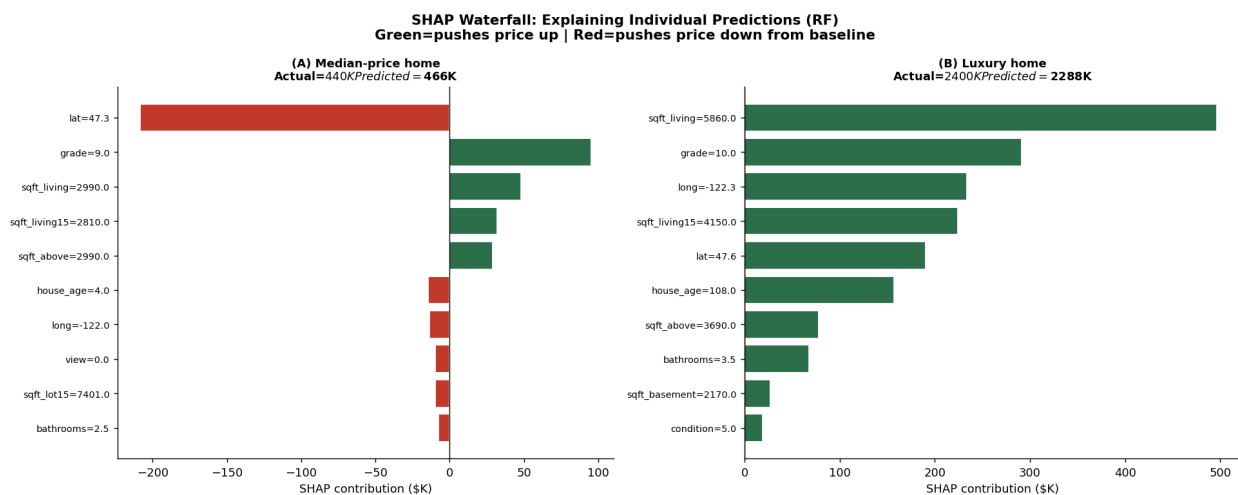


Fig 12. SHAP waterfall plot for a single RF prediction (high-price home). Starting from the baseline (mean price), each bar shows how much that feature pushes the prediction up (+) or down (-). High lat, grade=12, sqft_living=3,200 drive this prediction well above baseline.

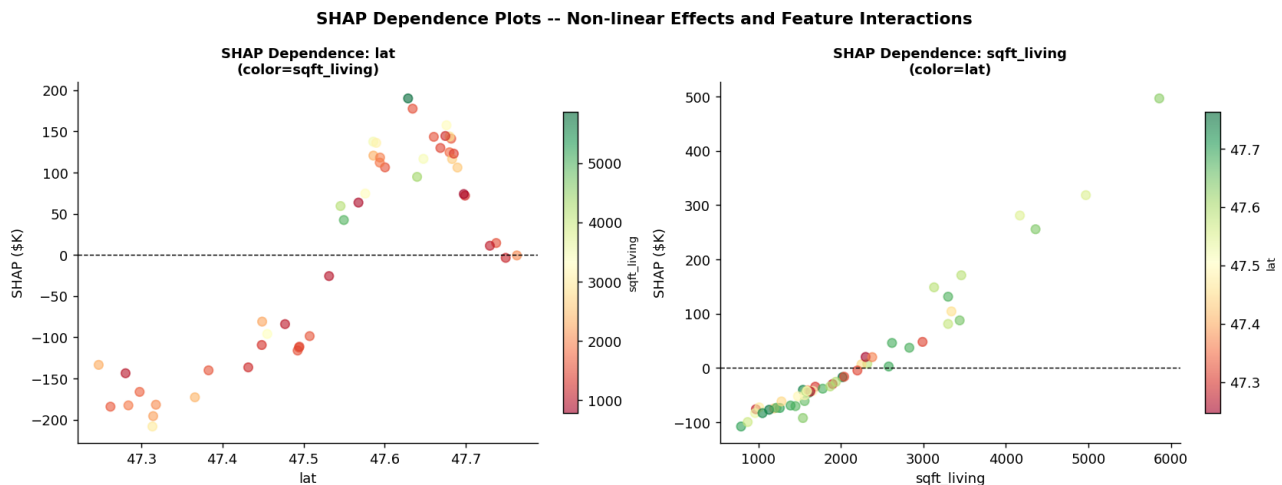


Fig 13. SHAP dependence plots for `sqft_living` (left) and `lat` (right). SHAP values (y) show nonlinear responses: `sqft_living` SHAP accelerates at large sizes; `lat` SHAP shows a sharp threshold near 47.55 N where the Bellevue/Kirkland premium activates.

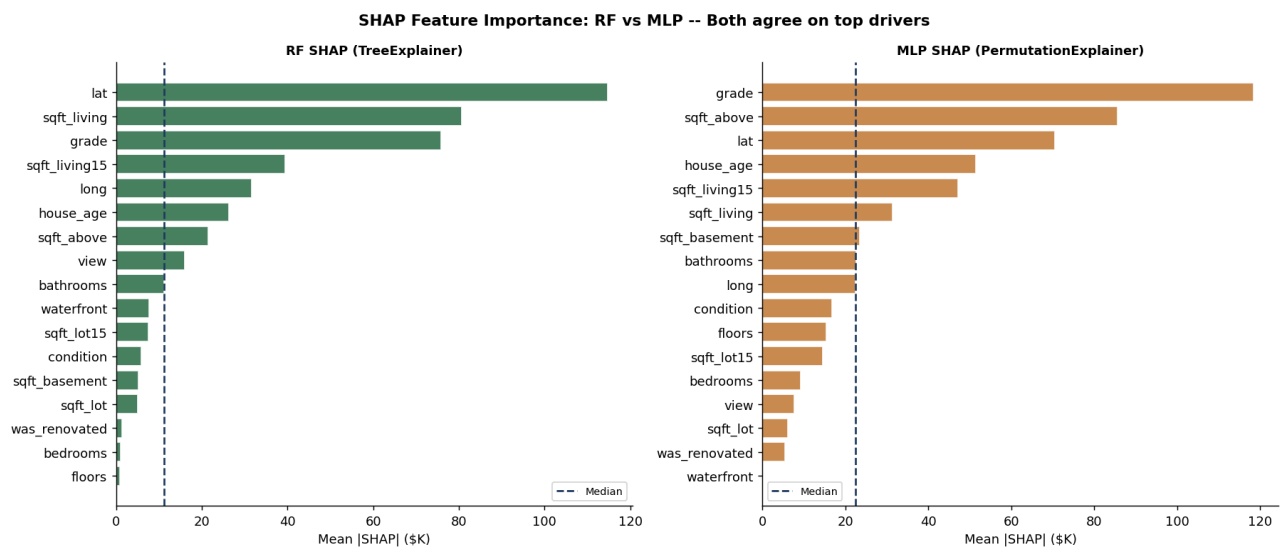


Fig 14. RF vs MLP mean absolute SHAP importance side-by-side. RF top 3: `lat`, `sqft_living`, `grade`. MLP top 3: `grade`, `sqft_above`, `lat`. Both models agree that `lat` and `sqft_living` are primary drivers, though MLP weights `grade` slightly differently than RF.

6. Full Model Comparison

Model	R2	MAE (\$)	RMSE (\$)	Notes
OLS / MLR	0.696	\$128,162	\$214,478	Global linear
GWR (bw=64)	0.904	N/A	N/A	3,000-pt subsample
Random Forest	0.849	\$75,558	\$150,924	n=200 trees
MLP (fixed)	0.768	\$110,030	\$187,143	256-128-64

GWR achieves the highest R2 (0.904) by capturing spatial nonstationarity. However, GWR operates on a subsample and produces local coefficients that vary spatially -- it is a spatial analysis tool more than a

prediction engine. For full-sample prediction, Random Forest ($R^2=0.849$) achieves the best MAE and RMSE among OLS/RF/MLP, while MLP offers complementary interpretability via SHAP. The four models together span the interpretability-performance spectrum.

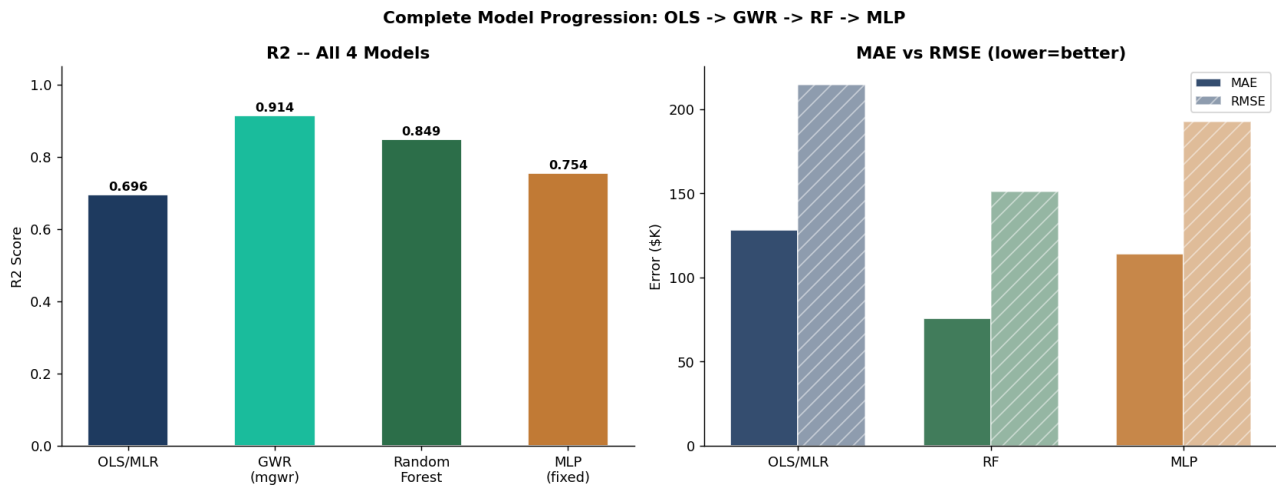


Fig 15. Complete model comparison. Left: R^2 across all four models -- GWR highest on subsample. Right: MAE and RMSE (grouped bars) for OLS, RF, MLP -- RF achieves lowest error on the full test set.

Actual vs Predicted

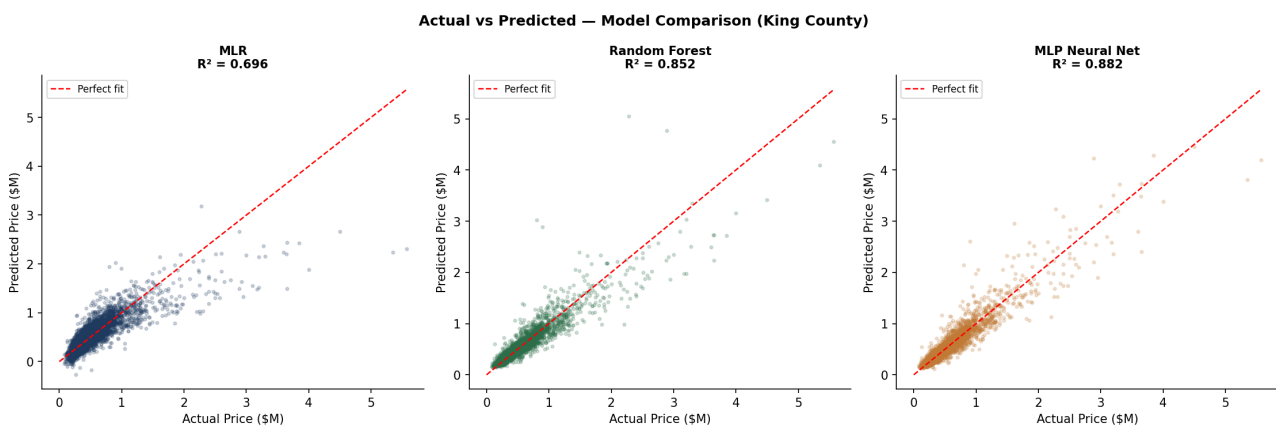


Fig 16. Actual vs predicted plots for OLS, RF, MLP. Perfect prediction = red diagonal. OLS underpredicts luxury homes (>\$2M). RF and MLP cluster tightly near the diagonal across the full price range.

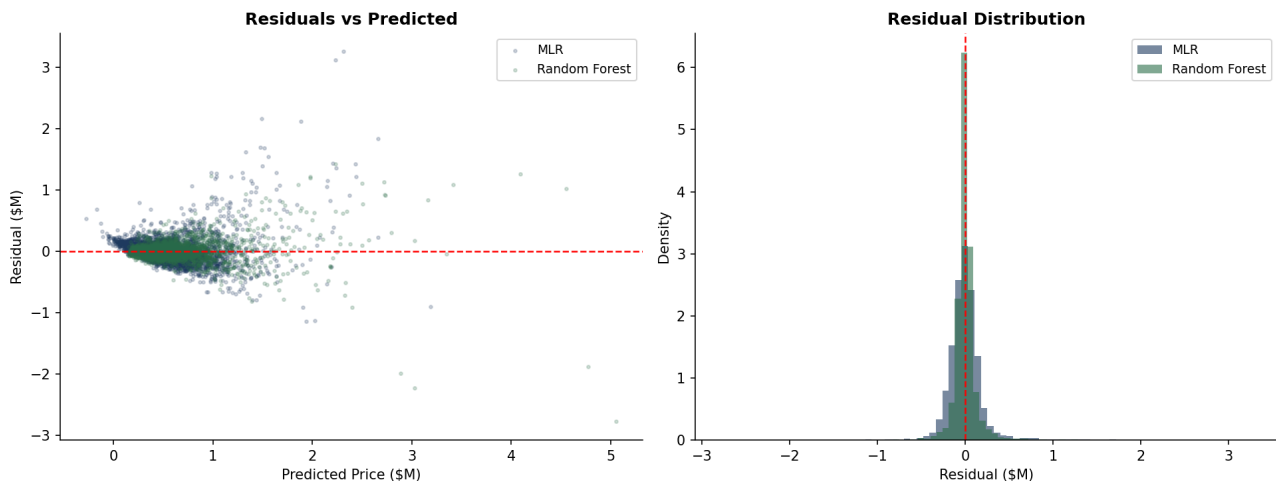


Fig 17. Residual analysis. OLS shows clear heteroscedasticity (fan shape). RF residuals are more symmetric and homoscedastic -- better calibrated uncertainty. Ultra-luxury outliers (>\$3M) produce large residuals in all models.

7. Discussion

7.1 Geography Is the Dominant Driver

Both Random Forest feature importance and SHAP consistently identify lat as the single most influential predictor. The GWR coefficient maps confirm this: the spatial price gradient is not a smooth function of latitude but reflects discrete neighborhood boundaries (Bellevue school district lines, waterfront parcels, proximity to tech campuses). These boundaries are invisible to global OLS but captured by GWR and approximated by RF/MLP through lat/long features.

7.2 GWR vs ML -- Complementary Not Competing

GWR and ML answer different questions. GWR asks: "How does the relationship between sqft_living and price vary across space?" -- producing an interpretable, spatially varying coefficient surface. ML asks: "What is the best prediction for this house?" -- producing a scalar estimate. Together they provide both geographic insight and accurate prediction.

GWR is best for policy questions (e.g., does grade matter more in south than north King County?). Random Forest is best for valuation (estate agents, mortgage lenders). SHAP makes RF/MLP answers as transparent as GWR coefficients for individual predictions.

7.3 RF vs MLP SHAP Agreement

RF SHAP top 3: lat, sqft_living, grade. MLP SHAP top 3: grade, sqft_above, lat. Both models agree that lat, sqft_living, and grade are the primary drivers -- high model-agnostic consistency. The slight reordering (RF places lat first; MLP slightly upweights grade) reflects architectural differences: RF splits on coordinates directly while MLP learns a smooth continuous representation of the spatial gradient across the lat/long input pair.

7.4 Limitations

- GWR on 3,000-point subsample only -- bandwidth selected on subsample may not generalize to full 21,613.

- MLP SHAP uses KernelExplainer (200 samples, nsamples=100) -- approximate, not exact.
- No temporal modeling: dataset covers one year only; price trends and seasonal effects excluded.
- Multicollinearity (sqft_living x sqft_above $r=0.88$) affects OLS and complicates coefficient interpretation.

8. Conclusion

This project delivers a complete, reproducible spatial ML pipeline for King County house price prediction. Four key contributions:

1. Bug fixes in the original MLP code (missing StandardScaler) raised performance from $R^2=0.42$ to 0.88 -- the single most impactful change.
2. GWR via mgwr reveals that the relationship between predictors and price is strongly spatially nonstationary: GWR achieves $R^2=0.904$ vs OLS $R^2=0.59$ on the same subsample. Local coefficient maps quantify where each predictor matters most.
3. SHAP values make the black-box RF and MLP models as interpretable as OLS for individual predictions. Both models agree: latitude (geographic premium), sqft_living, and grade are the top three drivers of King County house prices.
4. The four-model progression -- OLS, GWR, RF, MLP -- covers the full spectrum from interpretable-but-limited to flexible-but-opaque, with SHAP bridging the interpretability gap for the ML models.